

Exploratory Data Analysis (EDA) Project Report

Student: Vennala Priya

Dataset: SuperMarket Analysis (cleaned_supermarket.csv)

Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on a supermarket sales dataset to understand the data, identify patterns, detect missing values, analyze relationships among variables, and visualize important trends.

Tools Used

Python, Pandas, Matplotlib, Seaborn, and Pydroid 3.

Methodology

1. Loaded the cleaned_supermarket.csv dataset using Pandas.
2. Displayed the first five rows using head().
3. Checked dataset information using info().
4. Verified missing values using isnull().sum().
5. Generated statistical summary using describe().
6. Calculated correlations for numeric columns.
7. Visualized data using a correlation heatmap, histogram, box plot, and scatter plot.

Observations

- The dataset contains 17 columns with sales-related information.
- Numeric variables show different ranges and distributions.
- The heatmap highlights correlations between numerical features.
- The histogram shows the distribution of gross income.
- The box plot helps identify possible outliers in gross income.
- The scatter plot illustrates the relationship between gross income and customer rating.

Conclusion

The EDA process provided a clear understanding of the supermarket dataset through statistical summaries and visualizations. The analysis identified relationships among variables and prepared the dataset for further predictive analytics or machine learning tasks.